

Inversion Mechanical Counting and Mathematics

One Physical Unit of Itself, Partitioned Inward

From the Axiom of 1 to the Closure at $\sqrt{2^9}$,
and the Inward Reading from the Boundary Back

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Abstract

This paper establishes the mathematics and mechanics of Inverse Mechanical Counting. Physical applications belong to Inverse Spherical Dual Hemisphere Quantum Operational Mechanical Theory (ISDHQOM Theory) and are reserved for future work. The system here is built from a single physical unit, 1, partitioned inward by halving. The 1 is the largest unit in the system; no larger physical whole is introduced, and larger values appearing in the chain are mechanical readings of inward partition. A single parity rule governs the readings: at even depths the reading is rational, at odd depths the reading carries a factor of $\sqrt{2}$. Five active counts unfold nine active depths and reach a first closure at $\sqrt{2^9} = 16\sqrt{2}$, with structural boundary 512 and angular boundary $512\sqrt{2} \approx 724.077$, alongside the machined integer $720 = 5 \times 9 \times 16$. From the single first count, four views of the one whole are produced — square outward, sphere outward, square inward, sphere inward — without ever redoing the count. The mathematics is the reading; the physical unit is the thing read.

Scope of this paper. This paper establishes the mathematics and mechanics of Inverse Mechanical Counting. Physical applications — including cosmological, energy/mass, and quantum readings — belong to Inverse Spherical Dual Hemisphere Quantum Operational Mechanical Theory (ISDHQOM Theory) and are reserved for future work. The forward-work section near the end of this paper sketches where the count and mechanics will be applied next. Within this paper, the focus is the count, the math, and the mechanics.

The Single Axiom and Its Operation

Axiom of 1. There is exactly one physical unit. This unit is the complete reality. All its equivalent views are

$$1 = -1 = \sqrt{1} = \sqrt{2^0}.$$

The square root belongs to the unit itself; it is not introduced separately.

The Only Operation. Inward halving of the unit, read under parity:

- Even depth: the reading is rational.
- Odd depth: the reading carries a natural factor of $\sqrt{2}$.

From this single axiom and single operation, the entire system is generated by reading the one unit deeper inward. No other axioms or external quantities are required.

Partition does not create. Partition reveals.

1 The Master Table — The Single First Count

This one table is the source of all four views. We never redo the count.

c	d	$\sqrt{2^d}$	$\sqrt{2^d}$ decimal	how many (2^c)	rational ($2^c \times \text{even}$)	irrational ($2^c \times \text{odd}$)	decimal value
0	0	1	1	1	$1 \times 1 = 1$	—	1
1	1	$\sqrt{2}$	≈ 1.414214	2	—	$2 \times \sqrt{2} = 2\sqrt{2}$	≈ 2.828427
2	2	2	2	4	$4 \times 2 = 8$	—	8
2	3	$2\sqrt{2}$	≈ 2.828427	4	—	$4 \times 2\sqrt{2} = 8\sqrt{2}$	≈ 11.313708
3	4	4	4	8	$8 \times 4 = 32$	—	32
3	5	$4\sqrt{2}$	≈ 5.656854	8	—	$8 \times 4\sqrt{2} = 32\sqrt{2}$	≈ 45.254834
4	6	8	8	16	$16 \times 8 = 128$	—	128
4	7	$8\sqrt{2}$	≈ 11.313708	16	—	$16 \times 8\sqrt{2} = 128\sqrt{2}$	≈ 181.019336
5	8	16	16	32	$32 \times 16 = 512$	—	512
5	9	$16\sqrt{2}$	≈ 22.627417	32	—	$32 \times 16\sqrt{2} = 512\sqrt{2}$	≈ 724.077

Table 1: The master table. Five active counts ($c = 1, 2, 3, 4, 5$) unfolding nine active depths ($d = 1$ through $d = 9$), seated alongside the axiom at rest ($c = 0, d = 0$). Even depths give rational readings; odd depths give irrational readings. The how-many column is 2^c , the count's domain count. The decimal column shows what each reading equals as a number.

Count 5 is the first closure. The chain reaches $32 \times 16 = 512$ on the rational side and $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$ on the irrational side at $c = 5, d = 9$. Everything past this point is a repeat at larger scale — the same five-count structure unfolding again, with no new mechanics introduced. The single first count is the whole system.

How the Square Root Operates in Inverse Mechanical Mathematics

In this system the square root is not an added symbol or external constant. It is the natural mechanical signature that appears when the single unit is partitioned inward under the parity rule.

When the unit 1 is halved repeatedly, after an even number of halvings all accumulated factors remain rational (pure powers of 2); after an odd number of halvings the accumulated factors necessarily produce a reading that carries $\sqrt{2}$. Thus $\sqrt{2}$ emerges directly as the parity marker of odd depths. It is what the unit naturally becomes when read at odd resolution.

This is inverse mathematics: we do not start with $\sqrt{2}$ and build outward. We start with the single unit 1 and move inward. The square root appears mechanically as the system alternates between rational (square) and irrational (diagonal/sphere) readings of the same physical object.

The master table demonstrates this cleanly: even depths yield the rational column (ending at 512), odd depths yield the irrational column (ending at $512\sqrt{2}$). The entire structure — including the machined seat at 720 — is produced without ever leaving the original unit or adding new objects.

The square root therefore operates here mechanically as the geometric link between even and odd partitions of a single whole.

The Standard Square Root and the Mechanical Square Root — The Same Operation

The standard textbook square root function $f(x) = \sqrt{x}$ is defined as a rule: it finds the value that, when multiplied by itself, returns the original number. It is described as the inverse of squaring, the root factor of a non-negative real, the operation that reduces magnitude for $x > 1$, and the geometric mapping from the area of a square to its side length.

The mechanical square root in this paper is not a different operation. It is the same operation, shown at the level of the mechanism that produces it. Where the standard definition states the rule, the inward halving of one physical unit *produces* the rule directly. The four standard properties land on the master table without modification:

Standard square root (the rule)	Mechanical square root (the mechanism)
Undoes squaring: if $y = x^2$, then $\sqrt{y} = x$.	Even depth and odd depth alternate by parity. The square root of an even-depth reading lands cleanly on the previous odd depth: $\sqrt{2^8} = 2^4 = 16$, and squaring 16 returns $2^8 = 256$. The inverse-of-squaring rule <i>is</i> the parity ladder.
Finds the root factor: $\sqrt{a}$ is the y such that $y \times y = a$.	The depth ladder <i>walks</i> the root factor. Each odd depth is the square root of the next even depth: $\sqrt{2}$ at $d = 1$ pairs with 2 at $d = 2$; $2\sqrt{2}$ at $d = 3$ pairs with 4 at $d = 4$; and so on. The pairing is structural, not derived.
Reduces magnitude for $x > 1$: $\sqrt{x} < x$.	Each halving reduces the unit toward smaller resolutions. The parity flip at odd depths is what carries the $\sqrt{2}$ factor that links each rational level to the irrational level above it. The same reduction, read inward instead of as an external function.
Maps area to side length: the square root of the area gives the side.	The square boundary is 512. The sphere boundary is $512\sqrt{2}$. The diagonal of a square with side 512 is $512\sqrt{2}$. The $\sqrt{2}$ ratio between the two boundaries <i>is</i> the side-to-diagonal relationship, recovered at closure.

Table 2: The standard square root and the mechanical square root are the same operation. The standard definition states the rule; the mechanical chain produces the rule directly from one axiom and one operation.

The standard definition tells you what $\sqrt{x}$ does. The mechanical chain shows you where $\sqrt{x}$ comes from: the axiom of 1, the single operation of inward halving, and the parity rule on the count ladder 0, 1, 2, 3, 4, 5 produce the square root mechanically as the link between even and odd partitions of the same physical unit. No function is imposed. The mechanism is the function.

The Axiom Extended by One Partition

The axiom at rest is $\sqrt{2^0} = 1$. When the chain is unfolded inward and read at the boundary, the very same axiom returns one partition deeper. The closure ratio of the sphere boundary against the square boundary is

$$\frac{724.077}{512} = \frac{512\sqrt{2}}{512} = \sqrt{2} = \sqrt{2^1}.$$

This is not a new quantity. It is the axiom of 1 extended by one mechanical partition — the same single unit, now seen at depth 1 instead of depth 0. The chain begins at $\sqrt{2^0}$ and the boundary returns $\sqrt{2^1}$. What lives between them are the views the partition reveals.

2 Part One — The Count-by-Count Unfolding

This section walks the chain one count at a time. Each active count unfolds one or two local depths. For each depth we state the exponent, the parity (even or odd), the reading forced by the parity rule, the value, and the number of domains active at that count.

The Axiom Seat and the Active Chain

Before walking the chain, one distinction must be fixed once and for all. The labels $c = 0$ and $d = 0$ are not a count and a depth produced by the chain — they are the axiom itself. No partition has been performed at $c = 0$, and no new depth has been produced at $d = 0$. The axiom is the unit at rest; it is the starting condition, not mechanical work.

The mechanical work of the chain is therefore done by:

- Five active counts: $c = 1, 2, 3, 4, 5$ — the five halvings.
- Nine active depths: $d = 1, 2, 3, 4, 5, 6, 7, 8, 9$ — the nine new depths produced by those halvings.

This gives the active product $5 \times 9 = 45$, the base mechanical unit that will machine the integer 720 later in the paper.

The axiom seat sits alongside the active chain as the unit at rest. When this paper labels $c = 0$ or $d = 0$ in a table, it labels the axiom seat — not a count or a depth that the chain has produced. All mechanical statements in the paper refer to the five active halvings and the nine active depths.

Count 0 — The Axiom at Rest

The axiom is the unit at rest. No partition has been performed, so there is exactly one domain: the whole of 1.

- Axiom seat labelled $c = 0, d = 0, \sqrt{2^0} = 1$.
- Exponent 0 is even; reading is the rational unit 1.
- Domains: 1.

The axiom seat is a rational singular reading — one unit, one domain — but it is not an active count or an active depth. It is the axiom of 1 read on the exponent ladder, seated underneath everything the chain will produce.

Label c	Label d	$\sqrt{2^d}$	Value	Parity → Reading
0	0	$\sqrt{2^0}$	1	even → rational

Table 3: The axiom seat at $c = 0, d = 0$. Unit at rest, no partition.

Count 1 — The First Active Halving at Depth 1

Count 1 performs the first partition of the axiom. The whole of 1 is halved into two domains. Count 1 owns the first active local depth on the exponent ladder: $\sqrt{2^1}$.

- Count 1 owns local depth $\sqrt{2^1}$.
- Exponent 1 is odd, so by the parity rule the reading is irrational.

- Reading: $\sqrt{2} \approx 1.414214$.
- Domains: 2.

Count 1 is an irrational singular: one depth, one reading, two domains. This is the first place on the ladder where $\sqrt{2}$ appears, and it appears because exponent 1 is the first odd exponent.

Count c	Depth d	$\sqrt{2^d}$	Value	Parity $\rightarrow$ Reading
1	1	$\sqrt{2^1}$	$\sqrt{2} \approx 1.414214$	odd $\rightarrow$ irrational

Table 4: Count 1 unfolds local depth $d = 1$. Irrational singular.

Count 2 — The First Dual, Depths 2 and 3

Count 2 partitions again. From this count forward, every count owns two consecutive local depths, because the axiom keeps being halved and the exponent ladder keeps advancing by two whole numbers per partition.

Count 2 owns local depths $\sqrt{2^2}$ and $\sqrt{2^3}$. One of those exponents is even (2) and the other is odd (3), so by the parity rule count 2 carries both a rational reading and an irrational reading at the same count step. This is the structural definition of a dual count: it is forced by owning one even and one odd local depth at the same time.

- Count 2, depth $\sqrt{2^2}$, exponent 2 (even) — rational reading 2.
- Count 2, depth $\sqrt{2^3}$, exponent 3 (odd) — irrational reading $2\sqrt{2}$.
- Domains: 4.

Count c	Depth d	$\sqrt{2^d}$	Value	Parity $\rightarrow$ Reading
2	2	$\sqrt{2^2}$	2	even $\rightarrow$ rational
2	3	$\sqrt{2^3}$	$2\sqrt{2} \approx 2.828427$	odd $\rightarrow$ irrational

Table 5: Count 2 unfolds local depths $d = 2$ and $d = 3$. First dual count.

Count 3 — Dual, Depths 4 and 5

Count 3 partitions again. It owns local depths $\sqrt{2^4}$ and $\sqrt{2^5}$ — again one even exponent and one odd exponent.

- Count 3, depth $\sqrt{2^4}$, exponent 4 (even) — rational reading 4.
- Count 3, depth $\sqrt{2^5}$, exponent 5 (odd) — irrational reading $4\sqrt{2}$.
- Domains: 8.

Count 4 — Dual, Depths 6 and 7

Count 4 partitions again and owns local depths $\sqrt{2^6}$ and $\sqrt{2^7}$.

- Count 4, depth $\sqrt{2^6}$, exponent 6 (even) — rational reading 8.
- Count 4, depth $\sqrt{2^7}$, exponent 7 (odd) — irrational reading $8\sqrt{2}$.
- Domains: 16.

Count c	Depth d	$\sqrt{2^d}$	Value	Parity $\rightarrow$ Reading
3	4	$\sqrt{2^4}$	4	even $\rightarrow$ rational
3	5	$\sqrt{2^5}$	$4\sqrt{2} \approx 5.656854$	odd $\rightarrow$ irrational

Table 6: Count 3 unfolds local depths $d = 4$ and $d = 5$.

Count c	Depth d	$\sqrt{2^d}$	Value	Parity $\rightarrow$ Reading
4	6	$\sqrt{2^6}$	8	even $\rightarrow$ rational
4	7	$\sqrt{2^7}$	$8\sqrt{2} \approx 11.313708$	odd $\rightarrow$ irrational

Table 7: Count 4 unfolds local depths $d = 6$ and $d = 7$.

Count 5 — Dual, Terminal Depths 8 and 9

Count 5 is the terminal count. It partitions one last time and owns local depths $\sqrt{2^8}$ and $\sqrt{2^9}$. When count 5 unfolds the odd local depth $\sqrt{2^9}$, the full active exponent ladder 1 through 9 has been laid down and the chain is complete.

- Count 5, depth $\sqrt{2^8}$, exponent 8 (even) — rational reading 16.
- Count 5, depth $\sqrt{2^9}$, exponent 9 (odd) — irrational reading $16\sqrt{2}$.
- Domains: 32.

The terminal pair of the chain is $(16, 16\sqrt{2})$. The terminal exponent is 9. The five active halvings have unfolded all nine active depths; the axiom at $d = 0$ remains at rest beneath them.

Count c	Depth d	$\sqrt{2^d}$	Value	Parity $\rightarrow$ Reading
5	8	$\sqrt{2^8}$	16	even $\rightarrow$ rational
5	9	$\sqrt{2^9}$	$16\sqrt{2} \approx 22.627417$	odd $\rightarrow$ irrational

Table 8: Count 5 unfolds the terminal local depths $d = 8$ and $d = 9$. Terminal pair: $(16, 16\sqrt{2})$.

3 Part Two — Counting Inside a Count

Every count has its own mechanics. Count 0 has them. Count 1 has them. Counts 2, 3, 4, and 5 each have them. The coefficient at any count's depth is itself a unit, and that local unit partitions inward by the same halving operation under the same parity rule. The chain inside a count mirrors the master chain exactly: five active local counts, nine active local depths, terminal at the local boundary.

This section shows the structure with one worked example. Count 2 is used because it is the first dual count, with a clean rational coefficient (2) and a clean irrational coefficient ($2\sqrt{2} \approx 2.828427$). The same construction applies inside every other count. Nothing about the example is special to count 2; what is shown is the operational mechanics that every count carries.

The Local Unit Inside Count 2

Count 2 owns master depths $d = 2$ (rational, $\sqrt{2^2} = 2$) and $d = 3$ (irrational, $\sqrt{2^3} = 2\sqrt{2} \approx 2.828427$). It has 4 master domains. The two coefficients 2 and 2.828427 are the master-level readings of count 2.

Inside count 2, those same two coefficients become the local “1 of itself.” The local rational unit is 2. The local irrational unit is 2.828427. They are the local axiom seats, and the inward halving operates on each one in the same mechanical way the master chain operates on the original 1. The 4 master domains are the context; the local unit being counted inside is the coefficient itself.

Local Count Table Inside Count 2

The table below walks the local chain from local count 0 (the coefficient at rest) down to local count 5 (the local closure). Each row gives the local depth, the parity (and therefore the type), the piece size at that local depth, the number of pieces, and the total value — which stays conserved at 2 on the rational side and 2.828427 on the irrational side throughout. The local boundary at local count 5 closes the inside-count chain exactly as the master chain closes at $c = 5$.

local count	local depth	type	piece size (each)	how many	total value	note
0	0	rational (even)	2	1	2	local “1 of itself” = $\sqrt{2^2}$
0	0	irrational (odd)	2.828427 ($2\sqrt{2}$)	1	2.828427	local “1 of itself” = $\sqrt{2^3}$
1	1	rational	1	2	2	half of 2
1	1	irrational	1.41421356 ($\sqrt{2}$)	2	2.828427	half of 2.828427
2	2	rational	0.5	4	2	0.5×4
2	3	irrational	0.70710678 ($\sqrt{2}/2$)	4	2.828427	0.7071×4
3	4	rational	0.25	8	2	0.25×8
3	5	irrational	0.35355339 ($\sqrt{2}/4$)	8	2.828427	half of 0.7071
4	6	rational	0.125	16	2	continuing halving
4	7	irrational	0.176776695 ($\sqrt{2}/8$)	16	2.828427	—
5	8	rational	0.0625	32	2	local rational closure
5	9	irrational	0.088388348 ($\sqrt{2}/16$)	32	2.828427	local sphere closure

Table 9: The local count table inside Count 2. The local rational unit (2) and the local irrational unit ($2.828427 = 2\sqrt{2}$) are partitioned inward by the same halving operation under the same parity rule that governs the master chain. Five active local counts unfold nine active local depths, terminating at the local boundary.

What This Worked Example Shows

The local chain inside count 2 has the same shape as the master chain. Five active local counts, nine active local depths, and a local closure at local count 5. The total value stays conserved at the local unit (2 rational, 2.828427 irrational) at every local count: as the pieces get smaller, more pieces appear, but the local whole is preserved. *Partition does not create. Partition reveals.*

This is the operational mechanics every count carries. The same construction runs inside count 0 (with local unit 1), inside count 1 (with local unit $\sqrt{2}$), inside count 3 (with local units 4 and $4\sqrt{2}$), inside count 4 (with local units 8 and $8\sqrt{2}$), and inside count 5 (with local units 16 and $16\sqrt{2}$). Each count’s coefficient is its own “1 of itself,” and that local unit unfolds inward by the same five-and-nine structure that governs the master chain.

The mechanics is mechanically self-similar at every level: the master chain is one count and nine depths; each master count contains its own internal chain; each internal chain closes at its local boundary. The single first count is the whole system, and the single first count is reproduced inside every count of itself.

4 We Are 1 Unit — A Physical Unit of Itself, Counted by Inward Mechanical Partition

The 1 is the whole. It is a physical unit *of itself* — meaning the unit and the number are the same single object, as the axiom states ($1 = -1 = \sqrt{1} = \sqrt{2^0}$). There is no separation between “the physical thing” and “the number 1.” They are one whole, counted as one.

The largest unit in the system is 1. No larger physical whole than the original unit is introduced; larger values appearing in the chain are mechanical readings of inward partition. The chain only ever moves inward, into the unit, never outward beyond it. Every number that appears — $\sqrt{2}$, 2, $2\sqrt{2}$, 4, up through the terminal $16\sqrt{2}$, and every closure at 512, 720, and 724.077 — is the same single 1 read at a deeper inward partition of itself. The numbers look bigger than 1 only because they are operational readings (how many at this depth, multiplied by the depth coefficient). The whole being read is still the one 1.

The chain begins from 1. The label 0 is only the seat — the rest label — of that same 1. No number is introduced from outside. No number is “added in” from some external supply. This is not a system where numbers are being freely invented by addition. This is a system where 1 is the whole, and the visible numbers are the mechanical readings of that one whole as it is partitioned inward and read by parity.

The operation is inward unfolding. **Partition does not create. Partition reveals.** The halving does not subtract pieces from the whole; it shows the whole at a deeper resolution. What the mechanics provide are the views the partition makes visible at each depth and direction.

Accordingly, the label $c = 0$, $d = 0$ is not an additive zero. It is the seat label for the unit at rest — the axiom before any partition has been performed. The name is 0; the reading is 1. The seat is where 1 sits on the count and depth ladders before any mechanical work is done.

5 Why the Count Produces the Dual — Rational and Irrational

The master table is not a list of arbitrary rows. It is the inevitable shape that one physical unit takes when partitioned inward under the parity rule. The dual structure — rational paired with irrational at every active count — is forced by the halving itself. Reading down the master table makes this visible row by row.

$c = 0$: The Seat

The axiom seat is the unit at rest:

$$\sqrt{2^0} = 1.$$

No halving has been performed. There is one reading, and its exponent is even. There is no parity flip yet because there is no partition yet. The seat owns the rational side alone.

$c = 1$: The Singular Halving

The first halving is forced to do something the seat cannot: it must produce a partition. One halving cannot land back on the same parity — if it did, nothing would have changed and no partition would have occurred. The halving *is* the parity flip.

So $c = 1$ lands on $d = 1$, producing

$$\sqrt{2^1} = \sqrt{2}.$$

The exponent is now odd. The reading carries $\sqrt{2}$ for the first time. This is the singular halving — one count, one depth, one flip. And in doing so, it establishes the dual structure: the rational seat 1 above and the irrational reading $\sqrt{2}$ below are now paired across the partition.

The dual is not a choice. It is what one halving of a unit of itself is forced to produce.

$c = 2, 3, 4, 5$: The Dual Halvings

Once the parity structure is established at $c = 1$, every later count carries it. Each of $c = 2, 3, 4, 5$ owns two consecutive active depths — one even, one odd — so the dual is present at every count from this point forward:

- $c = 2$ owns $d = 2$ (rational, $\sqrt{2^2} = 2$) and $d = 3$ (irrational, $\sqrt{2^3} = 2\sqrt{2}$).
- $c = 3$ owns $d = 4$ (rational, $\sqrt{2^4} = 4$) and $d = 5$ (irrational, $\sqrt{2^5} = 4\sqrt{2}$).
- $c = 4$ owns $d = 6$ (rational, $\sqrt{2^6} = 8$) and $d = 7$ (irrational, $\sqrt{2^7} = 8\sqrt{2}$).
- $c = 5$ owns $d = 8$ (rational, $\sqrt{2^8} = 16$) and $d = 9$ (irrational, $\sqrt{2^9} = 16\sqrt{2}$).

This is what the dual is: one count carrying both parities, rational paired with irrational, at every active step from $c = 2$ down to closure.

The Active Depth Tally

At closure ($c = 5$, $d = 9$) the nine active depths $d = 1$ through $d = 9$ are fully unfolded, seated alongside the axiom at rest, and the chain has produced the terminal pair $(16, 16\sqrt{2})$. The tally of active depths reads directly off the count structure:

$$\underbrace{1}_{\text{singular } c=1} + \underbrace{2}_{\text{dual } c=2} + \underbrace{2}_{\text{dual } c=3} + \underbrace{2}_{\text{dual } c=4} + \underbrace{2}_{\text{dual } c=5} = 9 \text{ active depths.}$$

One singular count plus four dual counts gives nine depths exactly. Five active counts, nine active depths, and the active product $5 \times 9 = 45$ all come from the same structure: one halving establishes the dual, and the dual propagates four times to closure.

The rational column closes at $32 \times 16 = 512$. The irrational column closes at $32 \times 16\sqrt{2} = 512\sqrt{2} \approx 724.077$. They close together, paired, because that is what the dual is — the rational reading and the irrational reading of the same single unit, partitioned inward to the same depth.

6 The Four Key Numbers

Four numbers carry the entire boundary structure of the chain. All four come directly from the master table — nothing new is introduced. Their pedigrees differ, and stating each pedigree explicitly avoids confusion later.

- 512 is the *rational closure* of the master table at $c = 5$, $d = 8$ (the square boundary).
- ≈ 724.077 is the *irrational closure* of the master table at $c = 5$, $d = 9$, derived as $512 \times \sqrt{2}$ (the sphere boundary).
- 720 is the *machined seat*, derived from the active product $5 \times 9 \times 16$ (the rational landing point between the two boundaries).
- ≈ 514.90 is the *derived symmetric counterpart*, derived from $512 \times (724.077/720)$ (the small-lift counterpart of 720 on the square side).

Number	Exact Expression	Decimal	Pedigree
512	32×16	512	Master table rational closure (square boundary)
720	$5 \times 9 \times 16$	720	Machined seat (active product $\times$ rational 16)
≈ 724.077	$512 \times \sqrt{2}$	≈ 724.077	Master table irrational closure (sphere boundary)
≈ 514.90	$512 \times (724.077/720)$	≈ 514.90	Derived symmetric counterpart of 720 on the square side

Table 10: The four key numbers and their pedigrees. All four come from the single first count: 512 is the rational closure, 724.077 is the irrational closure, 720 is the machined seat between them, and 514.90 is the derived symmetric counterpart of 720 on the square side.

The Ratios That Connect Them

The four numbers are bound together by two ratios: one large lift from the square boundary up to the machined seat, and one small lift from the machined seat up to the sphere boundary. The same small ratio also lifts 512 to 514.90, giving the symmetric pair across the machined seat.

Square to machined seat.

$$\frac{720}{512} = 1.40625 \quad (\text{exactly } \frac{45}{32})$$

This is the upward machining ratio from the square boundary 512 to the machined seat 720. The exact fraction $45/32$ is itself a count signature: $45 = 5 \times 9$ is the active product, and $32 = 2^5$ is the count's how-many at $c = 5$.

Machined seat to sphere boundary.

$$\frac{724.077}{720} \approx 1.0056625$$

This is the small final lift from the machined seat 720 to the full sphere boundary 724.077. It is the residue between the rational machined integer and the irrational $\sqrt{2}$ -carrying closure.

The symmetric small lift above the square boundary.

$$514.90 \approx 512 \times \frac{724.077}{720}$$

The same small ratio, applied to 512, produces 514.90 — the derived symmetric counterpart of 720 on the square side. The structure is symmetric across the machined seat: 512 sits below 720 by the large machining ratio, and 514.90 sits above 512 by the same small ratio that lifts 720 to 724.077.

The two ratios multiply to the axiom's first extension. The large lift and the small lift are not independent. Together they reproduce the axiom of 1 extended by one partition:

$$\underbrace{\frac{720}{512}}_{45/32} \times \underbrace{\frac{724.077}{720}}_{\approx 1.0056625} = \frac{724.077}{512} = \sqrt{2} = \sqrt{2^1}.$$

The square boundary times $\sqrt{2}$ is the sphere boundary, and the machined seat is the rational landing point where the partition factors $\sqrt{2}$ into a rational piece ($45/32$) and a small irrational residue (≈ 1.0056625). The chain begins at $\sqrt{2^0}$ and the boundary returns $\sqrt{2^1}$. The four key numbers and the two ratios are all one count. Nothing has been added.

7 The Units — Master and the Four Locals

The Master Table above is the single source. The four local tables below are the inward readings of the four key numbers, machined directly from the Master — nothing added, nothing external.

512 Inward (Square Boundary)

c	local depth	compounding domains	piece name	each piece + exponents
0	0	1	one	1/512
1	1	dual	dual	1/256¹ 1/256²
2	2	dual + quadrant	quadrant	1/128¹ 1/128² 1/128³ 1/128⁴
3	3	... + octant	octant	1/64¹ ... 1/64⁸
4	4	... + hexadecant	hexadecant	1/32¹ ... 1/32¹⁶
5	5	... + duo-hexadecant	duo-hexadecant	1/16¹ ... 1/16³²

≈ 514.90 Inward (Derived Symmetric Counterpart)

c	local depth	compounding domains	piece name	each piece + exponents (approx)
0	0	1	one	1/514.90
1	1	dual	dual	1/257.45¹ 1/257.45²
2	2	dual + quadrant	quadrant	1/128.725¹ ... 1/128.725⁴
3	3	... + octant	octant	1/64.3625¹ ... 1/64.3625⁸
4	4	... + hexadecant	hexadecant	1/32.181¹ ... 1/32.181¹⁶
5	5	... + duo-hexadecant	duo-hexadecant	1/16.091¹ ... 1/16.091³²

720 Inward (Machined Angular Cycle)

c	local depth	compounding domains	piece name	each piece + exponents	angle
0	0	1	one	1/720	720
1	1	dual	dual	1/360¹ 1/360²	360
2	2	dual + quadrant	quadrant	1/180¹ ... 1/180⁴	180
3	3	... + octant	octant	1/90¹ ... 1/90⁸	90
4	4	... + hexadecant	hexadecant	1/45¹ ... 1/45¹⁶	45
5	5	... + duo-hexadecant	duo-hexadecant	1/22.5¹ ... 1/22.5³²	22.5

≈ 724.077 Inward (Sphere Boundary)

8 The Four Views of the One Whole

All four views — square outward, sphere outward, square inward, sphere inward — are read from the single master count. No new halvings, no new depths, and no external quantities are introduced. Direction of reading and parity selection produce the distinct views from the identical partitioned unit.

c	local depth	compounding domains	piece name	each piece + exponents (approx)
0	0	1	one	1/724.077
1	1	dual	dual	1/362.039¹ 1/362.039²
2	2	dual + quadrant	quadrant	1/181.019¹ ... 1/181.019⁴
3	3	... + octant	octant	1/90.510¹ ... 1/90.510⁸
4	4	... + hexadecant	hexadecant	1/45.255¹ ... 1/45.255¹⁶
5	5	... + duo-hexadecant	duo-hexadecant	1/22.627¹ ... 1/22.627³² (16√2)

Square Outward

Read the master table rational column outward from the axiom seat. Domains compound: $1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32$. The terminal rational reading is taken at full compounding. This view shows the orthogonal lattice structure of the unit — pure doubling at every count, all readings rational. The boundary is the square closure.

Sphere Outward

Read the master table irrational column outward from the axiom seat. The same domain compounding occurs ($1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16 \rightarrow 32$), but each odd depth injects the natural $\sqrt{2}$ factor required by parity. The terminal irrational reading at $d = 9$ gives the full sphere closure. This is the diagonal reading of the identical unit: the same partitions, now carrying their spherical signature.

Square Inward

Read from the rational boundary (512) back toward the axiom seat using the 512 inward table. The whole is subdivided at each local depth: one $\rightarrow$ dual $\rightarrow$ quadrant $\rightarrow$ octant $\rightarrow$ hexadecant $\rightarrow$ duo-hexadecant. Every piece remains rational. This view reveals the nested orthogonal structure inside the unit at successively finer resolutions.

Sphere Inward

Read from the irrational boundary (≈ 724.077) back toward the axiom seat using the ≈ 724.077 inward table. The identical subdivision pattern applies, but odd-local-depth pieces carry the $\sqrt{2}$ scaling visible at the terminal duo-hexadecant row ($16\sqrt{2}$ pieces). This view exposes the continuous diagonal structure inside the discrete partitions.

The master table is the sole source. Outward readings compound domains; inward readings divide the boundary. Square selections suppress the $\sqrt{2}$ factor (even parity only); sphere selections retain it (full parity). All four views coexist as different mechanical readings of the one partitioned 1.

9 Self-Similarity at Closure

Count 5 closes the first active chain at $d = 9$. The structure is mechanically self-similar: the next five counts ($c = 6$ to $c = 10$) repeat the identical pattern at the next scale. The new rational closure will sit at $512 \times 32 = 16384$, the new sphere closure at $16384\sqrt{2}$, and the next machined angular seat at $720 \times 32 = 23040$.

No new mechanics appear. Every subsequent cycle is the first count read at deeper global resolution. The single first count generates all higher structure by simple repetition of the same five-count, nine-depth seed.

10 The Angular Cycle as Machined Property

The integer 720 is not imposed; it is machined directly from the active product $5 \times 9 = 45$ multiplied by the terminal rational coefficient 16. In the 720 inward table this produces exact angular halving at each local depth:

- $c = 0$: 720
- $c = 1$: 360
- $c = 2$: 180
- $c = 3$: 90
- $c = 4$: 45
- $c = 5$: 22.5

These angles emerge as the natural rational landings of the inward square reading when the active mechanical product is seated at the terminal rational coefficient. The small residue between 720 and the sphere boundary is the irreducible spherical remainder after rational machining. The full sphere boundary is recovered exactly by re-applying the axiom's first extension $\sqrt{2}$.

11 Return to the Axiom

Reading any of the four boundaries fully inward returns exactly the original unit 1. The terminal subdivided pieces at local depth 5 compound (outward) or divide (inward) back to the axiom seat at $c = 0$, $d = 0$. The entire system is a closed mechanical loop: every path begins at 1 and returns to 1. **Partition does not create. Partition reveals.** The partitions do not create multiplicity; they reveal the internal structure of the single physical unit.

The mathematics is the reading. The physical unit is the thing read.

12 Forward Work — Within ISDHQOM Theory

The count, math, and mechanics established in this paper are the foundation. Their physical applications belong to Inverse Spherical Dual Hemisphere Quantum Operational Mechanical Theory (ISDHQOM Theory), reserved for future work. This section sketches where the count and mechanics will be applied next, framed within the Theory rather than asserted as physical fact in this paper.

Within ISDHQOM Theory, the singularity is mathematically modeled as the 1 of itself — the axiom, the whole physical unit before any partition, the starting state. The big bang is mathematically modeled as the first halving: the step from $c = 0$ to $c = 1$, the singular halving, the moment partition begins. At that moment the parity flip happens, the unit is read at odd depth, $\sqrt{2}$ enters, and the chain has started.

Within ISDHQOM Theory, the dual takes its physical form as decompression of energy and compression of mass — the same rational/irrational pair the count produces, modeled at cosmological scale. The pairs 512/514.90 and 720/724.077 from the chain's boundary structure carry this reading, and the four views of the one whole carry it across both directions of the partition.

This count, math, and mechanics will be used in our future work in mathematics, physics, and quantum mechanics. The framework here is the seed; the applications follow within ISDHQOM Theory.

Conclusion. From the single axiom of 1 and the single operation of inward halving under parity, the complete system — including rational/irrational duality, four directional views, mechanical self-similarity at closure, machined angular structure, and return to the unit — is generated without external quantities. The largest entity is always the original physical unit. All structure is inward reading of that one whole. *Partition does not create; partition reveals.* From this foundation, the same count and mechanics extend forward into mathematics, physics, and quantum mechanics within Inverse Spherical Dual Hemisphere Quantum Operational Mechanical Theory (ISDHQOM Theory), where the singularity is mathematically modeled as the axiom and the big bang as the first halving.